

Assignment 3: End-to-End ML Versioning

Part C: DVC Setup with Google Drive as Remote

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Repository: github.com/Munam32/MLOps

Remote: gdrive_storage

How Git and DVC split the work

Git	Stores code and small text files, including tiny <code>.dvc</code> pointer files that record each dataset's hash.
DVC	Stores the large files themselves: first in a local cache (<code>.dvc/cache</code>), then in a <i>remote</i> , here a Google Drive folder.
Remote	Where <code>dvc push</code> uploads data and <code>dvc pull</code> downloads it, the same way <code>git push</code> works for code.

C1 Install DVC with Google Drive support

Goal: install DVC and its Google Drive plugin.

`pip install "dvc[gdrive]"` installs DVC plus `dvc-gdrive`. One version clash had to be fixed: the pinned `pyOpenSSL` crashed with the installed `cryptography` library (module `'lib'` has no attribute `'GEN_EMAIL'`), so it was upgraded with `pip install --no-deps "pyOpenSSL==26.4.0"`. This step is noted in `requirements.txt`.

```
$ pip show dvc dvc-gdrive | grep -E '^(Name|Version)'  
Name: dvc  
Version: 3.67.1  
Name: dvc-gdrive  
Version: 3.0.1
```

C2 Initialise DVC

Goal: turn the Git repository into a DVC project, on `dev`, after Part A's first commit.

`dvc init` created the `.dvc/` folder and `.dvcignore`, committed as `d85dffc`.

```
$ git show --stat --format='%h %s' d85dffc  
d85dffc feat: setup dvc  
  
.dvc/.gitignore | 3 +++  
.dvc/config      | 4 ++++  
.dvcignore      | 3 +++  
3 files changed, 10 insertions(+)
```

C3 Register Google Drive as the default remote

Goal: point DVC at a Drive folder.

I created a folder in Google Drive and copied its ID from the browser URL, then ran `dvc remote add -d gdrive_storage gdrive://<FOLDER_ID>`. The `-d` flag makes it the default, so a plain `dvc push` uses it. The setting is saved in `.dvc/config`, which is committed so anyone who clones the repo can `dvc pull`.

```
$ dvc remote list  
gdrive_storage gdrive://10YsvNKN6_F3Htff1tR2YluxfCFKZVRo8 (default)  
  
$ cat .dvc/config  
[core]  
    remote = gdrive_storage  
['remote "gdrive_storage"']  
    url = gdrive://10YsvNKN6_F3Htff1tR2YluxfCFKZVRo8
```

C4 Sign in, without committing credentials

Goal: authenticate with Google and prove no login file reaches Git.

The first `dvc push` opened Google's sign-in page, but Google blocked DVC's built-in app ("This app is blocked"). I created my own OAuth client (Google Cloud project, Drive API enabled, consent screen with my account as test user, Desktop-app client ID) and gave it to DVC with `--local`:

```
dvc remote modify --local gdrive_storage gdrive_client_id / gdrive_client_secret
```

The next `dvc push` signed in successfully. Credentials end up in two places, and neither is in Git:

FILE	HOLDS	WHY IT ISN'T COMMITTED
<code>.dvc/config.local</code>	Client ID and secret	Listed in <code>.dvc/.gitignore</code> (line 1)
<code>~/ .cache/pydrive2fs/</code>	The Google sign-in token	Outside the project folder entirely

```
$ git status --short
?? report/screenshots/

$ cat .dvc/.gitignore
/config.local
/tmp
/cache

$ git check-ignore -v .dvc/config.local
.dvc/.gitignore:1:/config.local .dvc/config.local

$ git ls-files | grep -i -E 'config.local|token|credential|secret' || echo '(no credential files tracked)'
(no credential files tracked)

$ ls ~/.cache/pydrive2fs/
3391...(client ID folder, holds the sign-in token)
7107...(client ID folder, holds the sign-in token)
```

C5 Track the data and push it to Drive

Goal: version the data and model with DVC, then upload them.

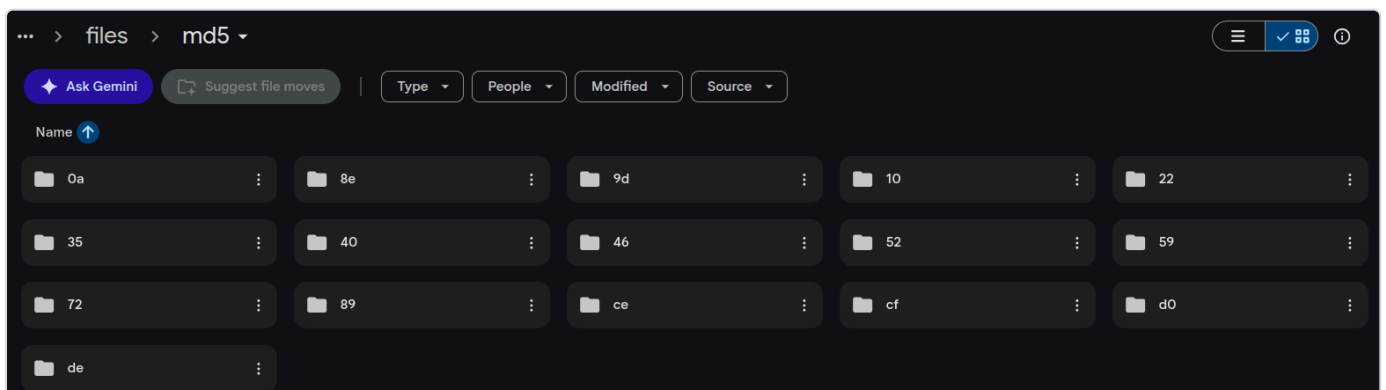
`dvc add data/raw data/processed models` hashed the three folders, copied them into the DVC cache, and wrote one small pointer file each. It also added the folders to `.gitignore` so Git skips the large files. Only the pointers and `.gitignore` changes were committed (`b433fb3`). A pointer file is just a hash and a size:

```
$ git show --stat --format='%h %s' b433fb3
b433fb3 feat: track raw data, processed data, and models

.gitignore      | 3 +--
data/.gitignore | 2 ++
data/processed.dvc | 6 ++++++
data/raw.dvc    | 6 ++++++
models.dvc      | 6 ++++++
5 files changed, 21 insertions(+), 2 deletions(-)

$ git show b433fb3:data/raw.dvc
outs:
- md5: 0a2fc63869fef5432996d7110256447d.dir
  size: 54951020
  nfiles: 2
  hash: md5
  path: raw
```

Then `dvc push` uploaded the cached files to Drive. On Drive, DVC stores each file under `files/md5/`, in a folder named after the first two characters of its hash:



The Google Drive remote folder, viewed in the browser.

In short: Git holds the pointer files (a few bytes each). Google Drive holds the ~270 MB of actual data and model files. `git checkout + dvc pull` brings back any version. In Part D these three pointers were replaced by the pipeline's `dvc.lock`, which records the same hashes.